

BREAK INTO AEROSPACE

Embraer

Engineering Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Embraer is the world's third-largest civil aircraft manufacturer and Brazil's most consequential aerospace company — founded in 1969 by the Brazilian government to develop domestic aerospace capability, and now operating across commercial aviation (E-Jet E2 family), executive aviation (Phenom, Praetor), defence (KC-390, Super Tucano), and urban air mobility (Eve eVTOL). Embraer is the only non-US, non-European OEM competing at global scale, certifying under ANAC with bilateral validation by EASA and the FAA, and operating primarily in Portuguese. This guide consolidates the preparation tasks, signal areas, and strategic context needed to compete effectively across all relevant engineering roles.

PART I · WHAT EMBRAER ACTUALLY LOOKS FOR

By Seniority Level

- **Intern / Trainee (*Estágio* / PEE Programme)** — Embraer's entry-level selection process is multi-stage and explicitly cultural. The process begins with a cultural fit test (*teste de fit cultural*), an English language test (non-eliminary at entry level), and a competency test — followed by a group phase with presentations and more tests, and culminating in a final interview where English is a priority. The cultural fit test is not a formality — it scores alignment with Embraer's named organisational pillars, and candidates are directly asked during the interview which pillar they most identify with. At intern level, Embraer is screening for three things simultaneously: academic technical grounding in an engineering discipline, genuine willingness to relocate to São José dos Campos (a mid-sized interior São Paulo city, not São Paulo itself), and evidence of multidisciplinary curiosity — because Embraer's integrated structure means engineers move between commercial, executive, and defence divisions throughout their careers.

The PEE (*Programa de Especialização em Engenharia*), Embraer's flagship early-career programme, is a direct pipeline from graduation to the company, run in partnership with ITA (Instituto Tecnológico de Aeronáutica). PEE is open to recent engineering graduates with advanced English, across a broad range of engineering disciplines, and combines a professional master's degree at ITA with applied project work at Embraer — with the possibility of being hired upon completion. For candidates eligible for PEE, it is the strongest possible entry into Embraer's engineering organisation.

- **Mid-Level (~3–8 years)** — At mid-level, Embraer expects engineers who can operate across the full product development lifecycle in a certification-governed environment. The primary regulatory framework is ANAC (Agência Nacional de Aviação Civil), Brazil's civil aviation authority, with bilateral validation by EASA and the FAA for market access. Engineers who have only worked with FAA or EASA as their primary regulator need to understand that ANAC's RBAC (Regulamento Brasileiro de Aviação Civil) is the governing standard from which bilateral validations flow. Mid-level candidates are also expected to demonstrate real engagement with Embraer's P3E (*Programa de Excelência Empresarial Embraer*) operational philosophy — embedded as a core business strategy since 2007, with the explicit formula: safety first, quality always, deliveries on time, all at best cost. Engineers who arrive from environments where Lean was aspirational rather than operational will need to demonstrate that they understand

value stream thinking, waste identification, and standard work — not just the vocabulary. *Careers in Aerospace*

- **Senior (10+ years, Specialist or Technical Leadership)** — Senior Embraer engineers carry two distinct expectations specific to the company. First, technical authority within a certification programme — the ability to prepare, defend, and close compliance findings with ANAC and with bilateral validation authorities, for products ranging from the E2 commercial jets to the Eve eVTOL. ANAC published final airworthiness criteria for Eve’s eVTOL design in late 2024, and Eve is working toward a means of compliance while conducting flight tests — representing an entirely novel certification frontier with no established playbook. Senior engineers who have navigated novel airworthiness criteria, rather than applying known standards to known products, are in the highest demand. Second, senior engineers are expected to be P3E champions — not just Lean practitioners but people who drive waste elimination at process level, lead kaizen events, and translate efficiency gains into measurable programme outcomes.

VT

Hands-On Signal Areas

Domain	Core Competencies Expected
English Technical Fluency	English is Embraer's primary language for international engineering — airline customer interfaces, EASA and FAA bilateral discussions, international supplier management, and global service centre coordination all happen in English. The final interview explicitly prioritises English level, and senior roles require technical fluency, not just proficiency.
ANAC / RBAC Familiarity	Embraer certifies all civil aircraft under ANAC's RBAC — RBAC 25 for large commercial aircraft, RBAC 23 for the Eve eVTOL. Engineers from FAA or EASA-only backgrounds need to understand the bilateral equivalence structure and where the regulatory frameworks diverge.
Lean / TPS Operational Experience	Embraer's Toyota partnership yielded a 50% reduction in manufacturing cycle for E-Jet wing production and a 17% decrease in total wing manufacturing time. Engineers who have driven quantified waste reduction, reduced takt time, or improved first-pass yield on a real production line are a strong signal. <i>Careers in Aerospace</i>
Multidisciplinary Engineering Experience	Embraer's integrated structure connects commercial, executive, and military divisions, with engineers regularly transferred between civil and defence departments. Engineers who have worked across more than one aircraft type, discipline, or regulatory framework are preferred over single-programme specialists. <i>European Space Agency</i>
Eve eVTOL & Urban Air Mobility Awareness	Eve's maiden flight in December 2025 and its ANAC certification process make urban air mobility Embraer's most forward-looking technical frontier. Engineers with backgrounds in electric propulsion, fly-by-wire control laws, novel certification methodologies, or urban aviation operations are specifically valued.
ITA / DCTA Institutional Connection	Embraer's engineering organisation is deeply connected to ITA (the Brazilian MIT equivalent), DCTA (the Aerospace Technical Centre), and the broader Vale do Paraíba aerospace ecosystem. Engineers who trained at or have collaborated with these institutions have cultural and technical pre-alignment with Embraer's engineering culture.
Portuguese Language or Acquisition Plan	While English is mandatory for international roles, Portuguese is the daily working language at every Brazilian facility. Non-Brazilian candidates who do not acknowledge this and articulate a realistic language acquisition plan are underestimating the daily professional reality.

How Embraer's Interview Format Differs from Every Other Company in This Series

- **Embraer's process is the most explicitly cultural of any company in this guide — and the cultural fit test is scored before you speak to a human.** The first selection stage consists of an online cultural fit test, an English test, and a competency test administered through the Gupy platform. Candidates who do not pass the cultural fit scoring threshold do not advance to the group phase regardless of their technical strength. The Gupy platform is Brazil's dominant recruitment technology for large companies

— not using it correctly is a process failure that has nothing to do with your engineering ability.

- **The group phase is a real selection event, not a formality.** The second phase involves group tests and a brief presentation of your academic or professional trajectory. It is observed for collaborative quality, communication clarity, and evidence of the cultural pillars. Candidates who approach it as a networking event rather than a selection event are misreading it.
- **The final interview asks about Embraer’s specific pillars — directly and explicitly.** Interviewers ask “which company pillar do you most identify with?” as a named, scored question. This is not a metaphorical values question — it requires knowing what Embraer’s specific pillars are, being able to connect one to your own professional experience, and giving a specific rather than generic answer.
- **English is actively evaluated in the final interview, not just checked on a form.** By the final interview stage, English is a priority — the interview itself may be conducted in English, or English fluency may be probed through specific questions. Candidates whose spoken English does not match their stated proficiency level are directly penalised.
- **For experienced professional hires, the format differs.** Direct hires at mid and senior level typically involve an HR screen, a technical interview with the hiring manager covering a resume deep-dive and domain questions, and potentially a panel including engineers from the relevant division. The format is conversational and the technical depth varies by division — deeper in structures and aerodynamics for commercial aircraft development, more operationally focused for customer support or service engineering roles.
- **The process timeline is 2–3 months for the multi-stage format,** and can be longer for experienced hires requiring relocation. This is characteristic, not exceptional.

Honest caveat on format variation: Embraer’s Brazilian operations are concentrated in the São Paulo interior (São José dos Campos, Gavião Peixoto, Taubaté, Botucatu). International operations — executive aviation service centres in Melbourne FL, Fort Lauderdale FL, Nashville TN; and engineering offices in the Netherlands, Portugal, Singapore — have different interview cultures and language dynamics. The multi-stage Gupy process applies primarily to Brazilian roles. International hires at Embraer Executive Jets or at international service centres follow a more conventional local-market process. If you are interviewing for a role outside Brazil, verify the specific process through the local HR contact.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Early Career (*Estágio / PEE / Trainee*)

1. Learn Embraer’s Named Organisational Pillars and Prepare a Specific Story for the One You Most Identify With 3 h

Skill: Cultural fit articulation aligned to Embraer’s scored framework

Objective: Interviewers directly ask “which company pillar do you most identify with?” as a named interview question — and the cultural fit test preceding the interview already scored your alignment against these pillars online. A candidate who arrives not knowing what the pillars are, or who gives a generic answer about “innovation” or “teamwork” without connecting to Embraer’s specific language, is demonstrating they have not engaged with the company. Embraer’s pillars include: People, Innovation, Sustainability, Customer Satisfaction, and Safety — each has a specific operational meaning at Embraer.

Done Signal: You can name all of Embraer's pillars, explain what each means in practice at an aerospace company, and deliver a 90-second story connecting one pillar to a real experience — where the pillar is evident in the content of the story, not just stated at the end.

3 h

2. Demonstrate English at Working Technical Fluency — Specifically by Practising Technical Topics in English

4–6 h

Skill: Technical English proficiency under interview conditions

Objective: The final interview stage explicitly prioritises spoken English level — and for PEE candidates, advanced English is a stated eligibility requirement. Embraer operates in over 50 countries, and engineers who cannot communicate in English cannot interface with the airline customers, regulatory authorities, international suppliers, and overseas service centres that are part of daily engineering life at a company of this scale. Conversational English is not sufficient — technical English (explaining a design decision, describing a failure mode, responding to a certification finding) must be practised specifically.

Done Signal: You have delivered a 5-minute technical explanation of a project or concept you know well — in English, out loud, to another person — and received feedback on vocabulary, clarity, and fluency. You can explain three aerospace engineering concepts (e.g., fly-by-wire, structural fatigue, single-engine go-around performance) in English without reverting to Portuguese for technical terms.

Spread over multiple days.

3. Complete the Gupy Platform Profile Fully and Correctly — Including the Cultural Fit and Competency Tests Under Real Conditions

2–3 h

Skill: Platform compliance and assessment performance

Objective: The selection process begins with Gupy and includes cultural fit, English, and competency tests — all online and scored before any human reviews an application. Candidates who treat these as formalities, rush them, or complete them in distracting environments risk self-eliminating before their CV is ever read.

Done Signal: Your Gupy profile is 100% complete with no blank fields. You have completed the assessments in a quiet environment, on a stable connection, with a camera and microphone tested. You have taken at least one practice competency assessment (available through preparation platforms) in timed conditions to calibrate your pacing.

4. Prepare an Honest, Specific Answer to “Are You Open to Working On-Site in São José dos Campos?” — with a Realistic Relocation Plan If Applicable

1–2 h

Skill: Relocation readiness and professional self-awareness

Objective: São José dos Campos is approximately 90 minutes from São Paulo city — a commute that requires either living in São José or accepting a substantial daily journey. Embraer asks directly whether candidates are willing to work *presencialmente*, and ambiguous or non-committal answers are penalised. International candidates face an additional language and cultural adjustment. Willingness to work in-person is an explicit interview question — not a soft preference.

Done Signal: You have a direct, honest 30-second answer to the relocation question — either “yes, and here is my timeline and plan” or “I would need to understand the role’s flexibility” stated without evasion. If relocating from outside Brazil, you have researched the practicalities (neighbourhood, cost of living, Portuguese classes) enough to discuss them credibly.

5. Study Embraer's Current Active Programmes and Be Able to Describe Each at a Technical Level 4 h

Skill: Portfolio knowledge that connects your background to a specific Embraer product

Objective: Embraer engineers work across four business units — commercial aviation (E-Jet E2 family), executive aviation (Phenom, Praetor), defence (KC-390 Millennium, Super Tucano), and urban air mobility (Eve eVTOL). Entry-level candidates who cannot connect their academic background to a specific Embraer programme are giving a generic answer. The candidate who says “my undergraduate thesis on flight control systems directly relates to the fly-by-wire architecture Embraer validated on the E2 and is now proving on the Eve” is demonstrably more prepared than the one who says “I want to work on aircraft.”

Done Signal: For each of Embraer's four product families, you can state: the primary aircraft and its market segment, the certification authority (ANAC/RBAC), one technically specific feature that distinguishes it from competitors, and one engineering challenge currently active in that programme. You can deliver this for all four in under 8 minutes total.

6. Understand What the P3E Programme Is and How It Governs Engineering Decisions — with a Lean Example from Your Own Work 2–3 h

Skill: Lean / P3E cultural alignment with a real operational example

Objective: Embraer's Lean philosophy — branded as P3E — has been the core business strategy since 2007, with the explicit formula: safety first, quality always, deliveries on time, all at best cost. This is not aspirational language — it governs how trade-offs are made, how improvement ideas are surfaced and validated, and how engineering decisions are justified in production reviews. Entry-level candidates who can connect their own experience (lab efficiency improvements, project waste identified, process standardised) to Lean principles are demonstrating pre-alignment to how Embraer actually operates.

Done Signal: You can explain what P3E stands for and what its four-part operating formula means in engineering practice. You have one specific personal example — however small — where you identified and eliminated waste, standardised a process, or improved quality in a repeatable way. The example names the waste or inefficiency, the action taken, and the measurable outcome.

Source: Careers in Aerospace

Mid-Level (~3–8 years)

All entry-level tasks still apply. English must now be working technical fluency — not just conversational — and the P3E example must come from a professional context with quantified results.

7. Study ANAC's RBAC Framework and Understand How It Relates to FAR and CS — and Where They Diverge 5–6 h

Skill: Brazilian civil aviation regulatory framework operational literacy

Objective: Embraer certifies all its civil aircraft under ANAC's RBAC — RBAC 25 (large aeroplanes, equivalent to FAR Part 25 and CS-25), RBAC 23 (normal category, governing the Eve eVTOL certification), and RBAC 21 (certification procedures, equivalent to FAR Part 21). ANAC has bilateral airworthiness agreements with both the FAA and EASA, meaning ANAC Type Certification enables validation in both markets. Mid-level engineers from FAA or EASA-only backgrounds need to understand that ANAC is the originating authority, not a secondary one, and that knowing where RBAC deviates from the bilateral equivalents is a live engineering skill.

Done Signal: You can explain what RBAC 25 governs, how it differs structurally from FAR Part 25

(same origin document, some divergences in specific amendments and national supplements), what the ANAC-FAA and ANAC-EASA bilateral agreements cover, and what “means of compliance” means in the context of ANAC certification. For the Eve eVTOL, you can explain why RBAC 23 was selected as the certification basis and what that means for novel airworthiness criteria.

5–6 h

8. Prepare a Story About Applying Lean Thinking to Improve an Engineering or Production Process — with Quantified Results

2–3 h

Skill: P3E / Lean operational contribution with measurable impact

Objective: Embraer’s Toyota partnership delivered a 50% reduction in manufacturing cycle for E-Jet wing production and a 17% decrease in total manufacturing time — these are the benchmarks the P3E programme is delivering. Mid-level engineers are expected to be active contributors to P3E, not passive participants. The story must be specific: what waste was identified (overproduction, waiting, transport, over-processing, inventory, motion, defects), what tool was used (value stream mapping, 5S, standard work, kaizen, SMED), what changed, and what the before/after metrics were.

Done Signal: You have a story where you led or significantly contributed to a Lean improvement — the waste category is named, the tool used is specific, the change implemented is described, and the quantified result (time saved, defect rate, cycle time, yield) is stated. The improvement was sustained after you left the area or the project.

Source: Careers in Aerospace

9. Study ARP4754A and Understand How Embraer Applies It to Its Integrated Multidisciplinary Development Process

5 h

Skill: Integrated development process standard for civil aircraft systems

Objective: ARP4754A governs the development of civil aircraft systems and equipment — it defines how aircraft-level functions are allocated to systems, how Design Assurance Levels (DAL) are assigned, and how development activities are planned and verified. Embraer applies this standard across its commercial and executive aircraft development. Mid-level systems engineers who arrive not knowing how ARP4754A relates to RBAC 25’s certification requirements, or who confuse it with DO-178C (which governs only software within that framework), are demonstrating a gap in commercial aircraft development literacy.

Done Signal: You can explain ARP4754A’s scope, how it relates to RBAC 25, what the Functional Hazard Assessment process produces, how DAL allocation flows from aircraft function through system to item, and why independence requirements at DAL A and B affect how Embraer staffs its development teams. You can describe where DO-178C and DO-254 fit within this framework.

10. Prepare a Story Demonstrating Cross-Divisional or Cross-Discipline Work — Reflecting Embraer’s Integrated Engineering Structure

1–2 h

Skill: Multidisciplinary flexibility in an integrated aircraft company

Objective: Embraer’s integrated structure connects commercial, executive, and military divisions, with engineers regularly transferred between them depending on demand. This is a structural feature, not an occasional occurrence — Embraer explicitly values engineers who can operate across product families and disciplines. Mid-level candidates who have only worked within a single programme, a single discipline, and a single aircraft type are demonstrating a narrowness that does not fit Embraer’s operational model.

Done Signal: You have a story where you worked across at least two engineering disciplines, two aircraft types, or two organisational boundaries — and the cross-boundary work produced a specific

technical or programme outcome. The story names both disciplines, the technical tension at the boundary, and what changed because of your involvement.

1–2 h

Source: European Space Agency

11. Develop Working Knowledge of Embraer's Fly-By-Wire Architecture — and Be Able to Discuss the E2's Fifth-Generation System and Its Application to Eve

4 h

Skill: Fly-by-wire architecture as Embraer's central technical signature

Objective: Embraer's fifth-generation fly-by-wire system is proven on the E2 family and KC-390, and has been validated on the Eve eVTOL during its December 2025 maiden hover flight. Eve's maiden flight validated the integration of Embraer's fifth-generation fly-by-wire system, which is already proven on the E2, KC-390, and other models. Fly-by-wire is not just a technology at Embraer — it is a core engineering heritage. Mid-level engineers who can discuss flight control architecture, actuator redundancy, control law design philosophy, and how the same platform is adapted from a commercial jet to an eVTOL are demonstrating fluency with the company's deepest technical identity.

Done Signal: You can explain what a fly-by-wire system is and why it replaced mechanical linkages, describe the redundancy architecture of a civil fly-by-wire system at a conceptual level, explain what a control law is and what its design must achieve for certification, and describe one specific technical challenge that arises when adapting an existing FBW system to a novel vehicle category such as an eVTOL.

Source: VT

12. Prepare a Story About Working with an International or Multicultural Team — Including Navigating a Technical or Process Difference That Had a Cultural Dimension

1–2 h

Skill: International engineering collaboration appropriate to a global OEM

Objective: Embraer's customers span over 50 countries, its supply chain is global, and its certification activities involve ANAC, EASA, and the FAA simultaneously. Engineers who have only worked in nationally homogeneous environments will find the international dimension of Embraer work unfamiliar. The probe is not just whether you have worked with international colleagues, but whether you have navigated a genuine technical or process misunderstanding that had a cultural root.

Done Signal: You have a story where you worked with colleagues or partners from a different country, a genuine misunderstanding or difference in approach arose, and you addressed it through specific action — not just by “communicating better.” The story names the cultural dimension of the difference (different regulatory reference, different engineering education tradition, different documentation standard, different communication norm) and what you changed in your approach.

13. Understand Embraer's Executive Aviation Product Line and Its US Market Context — Specifically for Roles at Melbourne FL or Other US Service Centres

3 h

Skill: Executive aviation product and customer fluency for non-Brazilian Embraer operations

Objective: Embraer's US executive aviation operations — centred in Melbourne FL — serve customers operating Phenom 100/300 and Praetor 500/600 aircraft. These customers are high-net-worth individuals, charter operators, and fractional ownership companies with very different expectations from airline customers. Engineers and technical support staff at these locations deal with FAA Part 91/135 operators, NBAA standards, and US-specific configuration requirements. Mid-level candidates interviewing for Melbourne-area roles who arrive without knowing the Phenom 300 is the world's best-selling light jet, or what a Praetor 600's range and cabin configuration enables, are missing the basic product literacy the role requires.

Done Signal: You can describe the Phenom and Praetor product families, their respective market segments, the FAA certification basis for each, and the primary service needs that bring operators to an Embraer service centre. You can explain one technically specific feature of an executive aviation aircraft that differs from a commercial regional jet and why it matters for maintenance and support engineering. 3 h

Senior Level (10+ years, Specialist or Technical Leadership)

All mid-level tasks still apply. ANAC/RBAC knowledge (task 7) must be operational — you should have contributed to a certification evidence package reviewed by ANAC or its bilateral counterpart. P3E (task 8) must include leading a kaizen event or value stream improvement at programme level, not just participating in one.

14. Prepare to Walk Through a Complete Aircraft System or Sub-System Development Lifecycle — from Requirements Through ANAC Certification Evidence Closure 4–6 h

Skill: End-to-end development ownership in an ANAC-governed certification context

Objective: Senior Embraer engineers are certification thread owners — they carry a technical baseline from requirements through compliance demonstration and regulatory acceptance. The specific probe is not just “what did you build?” but “what did ANAC accept as evidence, and how did you get there?” This includes: the means of compliance selected, the test evidence package prepared, the ANAC review comments received and how they were dispositioned, and any deviation or exemption that was required and how it was justified.

Done Signal: For one aircraft system or subsystem you personally owned, you can answer questions at every development phase without notes — including the means of compliance statement, the most difficult ANAC (or FAA/EASA) comment received, how you dispositioned it, what the final compliance finding said, and what you would do differently at the requirements stage to simplify the compliance path.

15. Prepare a Technically Grounded Position on the Eve eVTOL Certification Challenge — Specifically on Novel Airworthiness Criteria and Means of Compliance 4 h

Skill: Novel certification methodology thinking for a frontier programme

Objective: ANAC published final airworthiness criteria for Eve’s eVTOL design in late 2024, and Eve is working toward a means of compliance while conducting flight tests. The Eve programme is the most technically novel certification challenge currently active at Embraer — there are no established precedents for all-electric distributed propulsion aircraft of this configuration, no proven means of compliance for many of the novel failure modes, and no historical fleet data to draw on. Senior engineers who can engage with this challenge technically — discussing what RBAC 23 can and cannot cover, what novel criteria ANAC has established, and what the engineering implications of a “lift and cruise” architecture are for redundancy and safety — are directly relevant to Embraer’s most forward-looking work.

Done Signal: You have a prepared 3-minute technical discussion of one specific certification challenge for the Eve eVTOL — the specific challenge is named (e.g., distributed propulsion failure case analysis, battery energy management under failure, control law certification for a novel flight envelope, vertiport infrastructure interface). Your position is technically grounded, names the relevant regulatory basis, and includes your view on how the compliance path might be structured.

Source: VT

16. Prepare Examples of Driving P3E / Lean Improvement at Programme Scale — with Documented Before-and-After Metrics 2–3 h

Skill: P3E leadership as a senior engineering function, not just participation

Objective: Embraer's Toyota partnership has produced measurable outcomes at programme scale — a 20% overall efficiency gain in wing manufacturing. Senior engineers at Embraer are expected to be P3E champions who drive improvements of this magnitude, not just execute their own work efficiently. The probe at senior level is not “have you used Lean tools?” but “have you led a value stream transformation, driven a standard work implementation across a team, or reduced a programme-level waste metric in a way that was documented and sustained?”

Done Signal: You have two stories where you led (not participated in) a Lean improvement at team or programme scale. Each story includes: the waste category identified, the Lean tool applied (value stream map, standard work, 5S, kaizen, SMED), the team size involved, the before and after metrics, and evidence that the improvement was sustained after the event.

17. Prepare a Story of Directly Interfacing with ANAC, EASA, or the FAA on a Technical Certification Matter — Including Delivering a Difficult or Novel Compliance Position 2 h

Skill: Regulatory authority interface at senior technical level

Objective: Senior Embraer engineers represent the company's technical position to regulators. The ability to prepare a compliance argument, present it clearly to an authority, receive technically challenging feedback, and resolve it with a formally documented outcome is a core senior function. The Eve programme in particular will require ongoing novel authority interface — every means of compliance for an unfamiliar failure mode is a direct technical dialogue with ANAC and its bilateral counterparts.

Done Signal: You have a story involving direct interface with a regulatory authority (ANAC, EASA, FAA, or equivalent) about a specific technical finding or compliance position. The story names the technical substance of the discussion, the authority's concern, how you prepared and delivered the Embraer position, and how the finding was formally closed — not resolved in a corridor conversation.

18. Build Quantitative Estimation Fluency for Aircraft-Level Performance Parameters — Without a Performance Deck or Simulation 4 h

Skill: Aircraft performance reasoning from first principles for senior engineering credibility

Objective: Senior Embraer engineers are expected to sanity-check analysis results, catch errors in supplier deliverables, and respond credibly to “does this number make sense?” in programme reviews. For commercial aircraft: fuel burn estimation from specific range and payload, payload-range trade-offs, approach speed implications of maximum landing weight, time-to-climb estimation. For executive aviation: range impact of cabin configuration changes, fuel burn at altitude. For eVTOL: hover power required versus battery energy density, range vs. hover time trade-offs for a given charge state. None of these requires a computer to estimate at order-of-magnitude level.

Done Signal: You have worked through five quantitative aircraft performance estimates in your specific domain — spoken aloud, with assumptions named before numbers — arriving at an order-of-magnitude answer with stated uncertainty. You have identified which assumption is most load-bearing in each estimate and how you would check it.

All Levels

19. Prepare a Specific, Technically Grounded Answer to “Why Embraer?” That Connects to a Specific Programme and to Brazil’s Aerospace Identity

1–2 h

Skill: Company-specific motivation that acknowledges Embraer’s national and institutional context

Objective: Embraer is not just a global aerospace company — it is a Brazilian national champion that grew from a government-founded entity into the world’s third-largest civil aircraft manufacturer. Embraer was founded in 1969 by the Brazilian government to develop domestic aerospace capability, initially supplying military aircraft before expanding into commercial aviation. Engineers who give a “Why Embraer?” answer without acknowledging this identity — who treat it like any other OEM — are missing something that genuinely matters to the organisation. The expected answer at every level connects technical interest in a specific programme to an honest appreciation of what Embraer represents in Brazilian and global aerospace.

Done Signal: You have a 90-second answer that names a specific Embraer programme and one technically specific feature of it, connects your engineering background to that programme, and includes one sentence acknowledging the specific character of Embraer as a company — its origin, its role in Brazil’s aerospace sector, or its unique position as the only non-US/European OEM at global scale — without sounding rehearsed or sycophantic.

Source: *Impact Pool*

20. Prepare an Honest Assessment of Your Portuguese Language Level and a Concrete Plan for Improvement If It Is Below B1

1 h + ongoing

Skill: Language self-awareness and preparation for a Portuguese-dominant daily environment

Objective: While English is mandatory for technical international work and senior roles, Portuguese is the daily working language at every Brazilian facility. Team meetings, lunch conversations, safety briefings, manufacturing floor interactions, design reviews with Brazilian suppliers, and informal technical discussions all happen in Portuguese. Non-Brazilian candidates who have not addressed this reality, or who plan to rely on English indefinitely, are underestimating the friction this creates and signalling limited commitment to the location. Brazilian candidates whose Portuguese is strong but technical vocabulary is underdeveloped in English face the inverse challenge.

Done Signal: You have an honest, calibrated self-assessment of your Portuguese level (none / A1–A2 / B1 / B2 / C1–C2). If below B1, you have enrolled in or scheduled structured lessons with a specific timeline toward B1 workplace fluency. If above B1, you have identified the technical vocabulary gaps relevant to your engineering domain (materials science, certification, manufacturing process terms) and a method for closing them. You can state this plan clearly and specifically in the interview without evasion.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Embraer pillars — story for most identified pillar	All	3 h
2	Technical English fluency practice — spoken	All	4–6 h
3	Gupy platform and assessments — full completion	All	2–3 h
4	On-site relocation answer — honest and specific	All	1–2 h
5	Four business unit programme knowledge	All	4 h
6	P3E / Lean — what it is and a personal Lean example	All	2–3 h
7	ANAC RBAC framework vs. FAR/CS — where they diverge	Mid+	5–6 h
8	Lean improvement story — quantified, named waste, tool	Mid+	2–3 h
9	ARP4754A and how it maps to RBAC 25 and Embraer’s DAL process	Mid+	5 h
10	Cross-division or cross-discipline work story	Mid+	1–2 h
11	Embraer FBW fifth-generation architecture and Eve application	Mid+	4 h
12	International multicultural collaboration story	Mid+	1–2 h
13	Executive aviation product line and US market context	Mid+	3 h
14	Full system development lifecycle through ANAC closure	Senior	4–6 h
15	Eve eVTOL certification challenge — technical position	Senior	4 h
16	P3E leadership — programme-scale improvement with metrics	Senior	2–3 h
17	Regulatory authority interface on certification matter	Senior	2 h
18	Aircraft performance quantitative estimation — domain	Senior	4 h
19	“Why Embraer?” — programme-specific + national identity	All	1–2 h
20	Portuguese self-assessment and concrete learning plan	All	1 h + ongoing

THE STRUCTURAL REALITY

Every other organisation in this guide is headquartered in the US or Europe, with regulatory frameworks (FAA, EASA), cultural norms, and engineering education traditions that most international candidates are already calibrated to. Embraer is headquartered in Brazil, certifies under ANAC, operates primarily in Portuguese, and has a national identity shaped by 50 years of building aerospace capability in a country that had to create the industry from scratch.

The candidates who succeed at Embraer are those who treat this not as a liability to manage but as a specific and interesting engineering and professional context to engage with. The ones who fail are those who treat Embraer as a slightly exotic version of Boeing or Airbus and are surprised that the regulatory framework, the language environment, and the cultural expectations are genuinely different from what they assumed. Walk into every Embraer interaction ready to demonstrate not just that you are a capable engineer, but that you have engaged honestly with what it means to work for the world’s only non-US, non-European OEM at global scale — and that you find that distinction genuinely interesting, not merely tolerable.

BREAK INTO AEROSPACE
